FROM ZERO TO RUNNING CODE

Python Development Environment Setup Guide

A concise, cross-platform setup for students who are new to Python.

1	Install Python Version 3.12 or newer
2	Choose an IDE VS Code or PyCharm
3	Create a project One folder per assignment
4	Add packages Use pip inside .venv
5	Run your code Shell or saved .py script

BEFORE YOU BEGIN

You need an internet connection, permission to install software, and about 20 minutes. If your computer is managed by an employer or school, request installation help before an assignment is due.

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1 Install Python and confirm pip

Install a current Python 3 release. The version shown by your terminal must be 3.12 or newer.

WINDOWS 10 OR 11

Go to python.org/downloads and install the Python Install Manager (the Microsoft Store version is also official).

- Open a new PowerShell or Windows Terminal window and run `python`. Accept installation of the latest stable Python 3 runtime if prompted.
- If a traditional installer offers Add Python to PATH, enable it. Keep pip selected.
- Verify with the commands below. On Windows, prefer `py` when the guide shows it.

macOS

Download the current macOS 64-bit universal2 installer from python.org/downloads and open the `.pkg` file.

- Use the standard installation. It supports both Apple Silicon and Intel Macs.
- Open the installed `Python 3.x` folder in Applications and run `Install Certificates.command`.
- Open a new Terminal window. Use `python3`; do not delete or change Apple's system Python.

LINUX

Open your distribution's Terminal and install Python, pip, and venv with its package manager:

```
# Ubuntu / Debian
sudo apt update
sudo apt install python3 python3-pip python3-venv

# Fedora / RHEL
sudo dnf install python3 python3-pip

# Arch Linux
sudo pacman -S python python-pip
```

Then run `python3 --version`. If it is older than 3.12, use your distribution's supported method for a newer release or contact course support. Do not replace the Python used by the operating system.

Verification

Windows

```
py --version
py -m pip --version
```

macOS / Linux

```
python3 --version
python3 -m pip --version
```

If pip is missing: Run `py -m ensurepip --upgrade` on Windows or `python3 -m ensurepip --upgrade` on macOS. On Linux, install the distribution's `python3-pip` package. Always use `python -m pip` rather than a bare `pip` command so packages go to the intended Python.

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2 Install one IDE

An IDE is the application where you write, run, and debug code. Python itself is installed separately.

OPTION A | VISUAL STUDIO CODE

Best for: a lightweight editor and students who want a simple, flexible workspace.

- Install VS Code from code.visualstudio.com/download.
- Open Extensions (the blocks icon), search for Python, and install the extension published by Microsoft.
- Choose File > Open Folder and select your CYBR312 project folder.
- Open the Command Palette, run Python: Select Interpreter, and choose Python 3.12+ or your project's `.venv`.
- Open a terminal with Terminal > New Terminal.

Official VS Code Python tutorial

OPTION B | PYCHARM

Best for: an all-in-one Python-focused workspace with project and environment tools built in.

- Install PyCharm from jetbrains.com/pycharm/download.
- The free core features are enough for this course; a paid Pro subscription is not required.
- Select New Project, choose your CYBR312 folder, and select a Python 3.12+ base interpreter.
- Let PyCharm create a project virtual environment, or select the existing `.venv` created on the next page.
- Open a terminal with View > Tool Windows > Terminal.

Official PyCharm installation guide

Confirm the IDE is using the right Python

THE INTERPRETER CHECK

- Open the IDE's integrated terminal.
- Run the version command for your operating system. It must say Python 3.12 or newer.
- After creating `.venv`, select that environment as the project interpreter.
- If the IDE shows a different version than your regular terminal, change the selected interpreter in the IDE.

> Do not install both unless you want to. VS Code and PyCharm can both complete the course work. Choose one and stay with it while you learn the workflow.

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3 Create a project and isolated environment

A virtual environment keeps this course's packages separate from other Python projects and from the operating system.

1. Create a folder such as `Documents/CYBR312`, then create a separate subfolder for each assignment. Open that assignment folder in your IDE.

2. In the IDE terminal, create a virtual environment named `.venv`:

Windows

```
py -m venv .venv
```

macOS / Linux

```
python3 -m venv .venv
```

3. Activate it. Your prompt should begin with `(.venv)` afterward:

Windows PowerShell	Windows Command Prompt	macOS / Linux
`.\.venv\Scripts\Activate.ps1`	`.venv\Scripts\activate.bat`	`source .venv/bin/activate`

PowerShell activation blocked? Use Command Prompt and its activation command, or select `.venv` as the IDE interpreter. Do not weaken a computer's security settings just to activate an environment.

4 Install required course packages

Activate .venv first. Then use the assignment's exact package names in place of the placeholders.

Parameterized command

```
python -m pip install <PACKAGE_1> <PACKAGE_2>
```

Replace every item in angle brackets; do not type the brackets.

Course examples

```
python -m pip install Pillow prettytable
```

Package names are not always the same as import names.

For an assignment-provided `requirements.txt` file:

```
python -m pip install -r requirements.txt
```

Install name -> import example

```
from PIL import Image
from prettytable import PrettyTable
```

Useful environment commands

```
python -m pip list
python -m pip install --upgrade pip
deactivate
```

Important: Install only the packages named by the course or assignment. Never use `sudo pip install`. Re-create `.venv` for a new project instead of installing into the operating system's Python.

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5 Use the Python shell

The terminal accepts operating-system commands. The Python shell accepts Python code and displays a >>> prompt.

Start on Windows

```
py
```

If `.venv` is active, use `python`.

Start on macOS / Linux

```
python3
```

If `.venv` is active, use `python`.

Try this after the >>> prompt appears:

```
>>> 2 + 3
5
>>> name = "CYBR312"
>>> print("Hello,", name)
Hello, CYBR312
>>> exit()
```

Prompt rule: The characters >>> are shown by Python. Do not type them. To exit, use `exit()`; Ctrl-D also works on macOS/Linux, and Ctrl-Z then Enter works in Windows Command Prompt.

Create and run a standalone script

- In the IDE, create a new file named `hello.py` inside your assignment folder.
- Enter the code below and save it. Make sure the filename ends in `.py`, not `.py.txt`.
- Open the IDE terminal in the same folder and run the command for your operating system.

```
# hello.py
import sys

print("CYBR312 Python setup is working!")
print("Python version:", sys.version)
```

Windows

```
py hello.py
```

macOS / Linux

```
python3 hello.py
```

SAFE FILE HABITS

- Keep one assignment per folder; save before running.
- Use short names with letters, numbers, and underscores, such as `week_2_hashing.py`.
- Do not name your file after Python or a library: avoid `python.py`, `PIL.py`, `prettytable.py`, `random.py`, or `json.py`.
- Submit the saved `.py` file requested by the assignment, not a screenshot of the editor.

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Troubleshooting and final check

'python' or 'py' is not recognized	Close and reopen the terminal. On Windows, use `py`; reinstall from python.org and enable PATH if offered. On macOS/Linux, use `python3`.
The version is older than 3.12	Install a newer Python 3 release, then select that interpreter in the IDE. Do not rely on an old operating-system Python.
ModuleNotFoundError	Activate `.venv`, then run `python -m pip install <PACKAGE_NAME>`. Confirm the IDE and terminal are using the same `.venv`.
pip installed the package, but imports still fail	Run `python -m pip --version` and `python -c "import sys; print(sys.executable)"`. Both should point into your project's `.venv`.
A script closes immediately	Run it from the IDE terminal instead of double-clicking the file. The terminal keeps output and error messages visible.
Permission or security error	Do not disable security controls. Use a user-owned folder such as Documents and ask your administrator or instructor for help.

READY FOR CYBR312 WHEN...

- [ ] Python reports version 3.12 or newer.
- [ ] The IDE is installed and its project interpreter is selected.
- [ ] The assignment folder contains an active `.venv`.
- [ ] Required packages install with `python -m pip`.
- [ ] The Python shell opens and exits normally.
- [ ] A saved `hello.py` script runs from the terminal.

OFFICIAL VENDOR RESOURCES

Python downloads
www.python.org/downloads/

Python on Windows
docs.python.org/3/using/windows.html

Python on macOS
docs.python.org/3/using/mac.html

Python on Unix/Linux
docs.python.org/3/using/unix.html

Installing packages with pip
packaging.python.org/en/latest/tutorials/installing-packages/

Virtual environments (venv)
docs.python.org/3/library/venv.html

VS Code Python tutorial
code.visualstudio.com/docs/python/python-tutorial

PyCharm installation
www.jetbrains.com/help/pycharm/installation-guide.html

When asking for help: Include your operating system, the exact command you ran, the complete error message, the result of the version check, and which IDE you chose. Do not send passwords, access tokens, or other sensitive information.

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